CancerOS Consolidated Predevelopment Audit

Audit ID: COS-AUD-001 | **Reconciliation baseline:** 1.0 | **Date:** 23 October 2026

Disposition: Foundation work may begin; feature integration is gated by the contract closures below. Full demo readiness is not established.

1 Decision and authoritative baseline

The twelve supplied PDFs form a broadly coherent research-prototype design. Keep the modular FastAPI backend, Next.js interface, PostgreSQL, private artifacts and asynchronous CPU workers. No architectural restart or rewrite of the twelve documents is needed. The main risks are inconsistent authority, incomplete executable contracts and unverified source/model dependencies.

This is a document-level audit, not a repository, dataset, provider-access or deployment verification. It establishes a concrete reconciliation for project-owner adoption; it does not claim that approval, implementation or release tests have already occurred. Original PDFs remain unchanged. Upon adoption, this report is the conflict-resolution overlay, with the following domain authorities:

Domain	Governing document	Boundary
Product intent and required journeys	PRD-001	Core outcomes; no clinical-use expansion
Architecture and service ownership	SPEC-007	Modular monolith and workers; earlier endpoint examples are superseded
Schema, payloads, REST and lifecycle	SPEC-008 plus this audit	Generate one OpenAPI/Pydantic/TypeScript contract set; no competing legacy routes
Data, imaging and simulation meaning	SPEC-003/004/005	Dataset/mask/time semantics; cannot override contract ownership or safety
AI processing and presentation	SPEC-006/009	Subject to SPEC-008 contracts and SPEC-010 restrictions
Safety and acceptance	SPEC-010/011	Safety restrictions and named evaluation gates govern enablement
Sequencing and literature	SPEC-012; RA-001 advisory	Schedule is a forecast; literature suggestions do not create mandatory scope

Canonical decisions for implementation:

- REST base: /api/v1/projects/{project_id}; /api/v1/me and health routes are explicit exceptions. Use SPEC-008 names, decimal-string quantities and typed time unions.

- State context is OBSERVED or SYNTHETIC in P0. Item provenance uses the five canonical values from SPEC-008/010; availability, QC and publication remain separate.

- A branch has one shared observed baseline plus at least three simulated follow-ups. Derived processing of synthetic content stays SYNTHETIC_GENERATED, with processing lineage.

- Source observations and published snapshot members are immutable. Corrections/retractions append records; branches stay pinned to their original baseline hash.

- P0 uses verified imported masks and deterministic measurements. Automated segmentation, learned forecasting, interpolation, HER2 inference and calibrated uncertainty remain disabled unless separately approved and evaluated.

- P0 requires grounded research summaries and evidence links; no particular external provider is assumed available. Source-span note extraction can be demonstrated on clearly labeled authorized/synthetic fixtures when real notes are absent.

- Five scenario families are required. The first vertical slice needs one verified real baseline; full declared demo readiness also requires a real repeated-imaging case. Five curated cases remains the final presentation target; a smaller delivered cohort must be declared reduced scope. The 100/500/2,000 figures are capacity targets, not first-build gates.

2 Contradictions resolved by this baseline

ID and source	Finding	Binding reconciliation
A01: SPEC-005 pp22-23; SPEC-006 pp30-32; SPEC-007 pp11-12; SPEC-008 pp8-13	Several incompatible REST catalogs and envelopes could generate different frontend/backend clients.	SPEC-008 governs. Do not implement /api/imaging, /api/ai, or /cases examples as parallel public APIs. UI browser routes are distinct.
A02: SPEC-004 p24; SPEC-010 p2	Synthetic derived measurements could be relabeled REAL_DERIVED.	SPEC-010 SAFE-02 governs: synthetic descendants retain synthetic origin; transformation metadata records how they were computed.
A03: SPEC-004 p25; SPEC-008 pp12-14; SPEC-009 p6	Simulator says publish after validation; later contracts require an explicit publication command.	Workers finish a validated run as ready. Authorized editor/admin publication makes it visible. Job success means durable output, not automatic public publication. Use idempotent guarded publication.
A04: SPEC-006 p32 versus SPEC-007 p17 and SPEC-010 p4	Meta interpretation and note extraction are P0 in one document; later design supports provider-neutral fallback and denied exports.	Preserve extraction/grounding behavior without hardcoding a vendor. Enable external inference only after policy/access verification. Label deterministic or saved results honestly; event-specific provider requirements remain unresolved.
A05: PRD p4; SPEC-004 p4; SPEC-008 p1	Four visits, four states and baseline counting are conflated.	Four branch positions = shared baseline + at least three synthetic follow-ups. Neither DCE phases nor branch copies count as real visits or independent patients.
A06: SPEC-004 p3/p18; SPEC-008 p6; RA-001 pp4-6	Unconstrained exposures, formal toxicity grading and literature experiments could expand P0.	Constrained scenarios only. Display source symptoms/labs without invented causal grading. Defer automatic CTCAE/RECIST classification, DCE explorer and perturbation jobs until their separate contracts/rules are adopted. Existing evidence/measurement drill-down remains in scope.

Source register

All twelve attached PDF copies were included in the cross-document review. Page references above are PDF page numbers. External scientific/licensing claims were assessed as dependencies rather than freshly reverified online; this audit adds no new clinical or legal claims.

Reviewed source set: PRD-001 (38 pages); SPEC-003 Clinical Data and Dataset (26); SPEC-004 Synthetic Longitudinal Data (36); SPEC-005 Medical Imaging and Digital Twin (28); SPEC-006 AI and Multimodal Intelligence (36); SPEC-007 Architecture (21); SPEC-008 Database and API Contracts (17); SPEC-009 UI/UX and Demonstration (17); SPEC-010 Safety, Privacy and Integrity (5); SPEC-011 Testing and Scientific Evaluation (7); SPEC-012 Roadmap and Demo (7); and COS-RA-001 Literature Addendum (7). The supplied (1).pdf copies are the audited versions.

3 Missing implementation details requiring early closure

These are not reasons to redraft the specification set. They are the coding agents' first contract/migration tasks, before dependent feature work merges.

ID	Gap and evidence	Required implementation decision and acceptance
C01 - UI read contracts	SPEC-009 pp14-15 explicitly proposes reads absent from SPEC-008; SPEC-012 p4 leaves closure until B24.	Move B24 contract closure ahead of frontend feature integration. Freeze capabilities/dictionaries, branch list, exact observation read, interpretation discovery and evaluation reads. Add job list only if Activity requires it. All new paths remain project-scoped and role/patient-authorized. Generate schemas, examples and clients together.
C02 - durable revision reconstruction	SPEC-008 pp4/8/11/14 promises as-of data_revision builds and stable pagination, but mutable heads and database snapshot isolation alone do not define durable history across worker requests.	Persist an append-only patient revision journal, ordered membership/head changes and retractions, or capture an immutable input manifest at enqueue. Specify which revision each list cursor pins. Test revision N after N+1 correction; never rebuild N from current heads.
C03 - relative-time root	SPEC-008 p3 requires every relative day to have an anchor and rejects cycles; p11 API uses anchor_state_id. A relative-only baseline has no fully specified root creation path.	Add an explicit patient timeline-origin identity independent of a state, with known reference meaning and time domain. Day zero may reference that origin; other offsets reference it or a valid acyclic anchor. Freeze API-to-storage mapping. Never invent an absolute date or self-reference the baseline.
C04 - snapshot assembly rules	SPEC-008 pp6/11/14 and SPEC-009 pp9-10 distinguish current/historical members but do not fully define selection windows, conflicting records, ties or carry-forward.	Version one assembly policy: exact source membership where known; no silent imputation or carry-forward; ambiguous values remain conflicts. Earlier imaging is historical context only. Record policy version and input IDs/hash. Test lab-only, late, same-day and conflicting records.
C05 - deployable security and configuration	SPEC-007/008 define authorization intentions; chosen identity integration, credentials, session/CSRF behavior, worker/storage permissions and concrete environment values are not supplied.	Record the selected auth/session contract and trusted claims; implement one authorization path. Pin runtime/package versions and configuration limits. Mock identity/provider only in isolated tests; no anonymous production bypass. Verify viewer/editor/admin and asset grants before staging exposure.
C06 - simulation and clinical dictionaries	SPEC-004 pp10-18 and SPEC-008 pp3/6/12 describe rules but do not supply executable analyte conversions, parameter distributions, eligibility rules, symptom scales and geometry tolerances.	Produce reviewed versioned config fixtures and JSON schemas for all five families, units, missingness and supported exposures. Unknown required pathology blocks that exposure. Use SPEC-011 p3 engineering tolerances as initial test rules, not clinical accuracy claims; freeze changes before evaluation.

4 Unresolved dependencies and release blocking risks

The attachments do not establish that the following evidence exists. Treat these as unverified dependencies, not confirmed external failures. Assign the named responsibility even if one person holds several roles.

Gate and owner	Evidence needed to close	What it blocks
D01 real baseline and mask - data/scientific owner	Actual source/release, allowed demo use, linkage, reviewed mask class, valid affine/units and overlay review; SPEC-003 pp20-24, SPEC-005 pp3-4/26	Real-data import publication and authentic anatomical measurement claims. Foundation coding may proceed on labeled engineering fixtures.
D02 genuine temporal case - data/scientific owner	At least two distinct real examinations, valid time relation, lesion correspondence and comparable measurement definitions	Full authentic longitudinal demonstration. Synthetic visits cannot close this gate. Registration is required for spatial overlays, not automatically for separate comparable scalar volumes.
D03 source rights/privacy - project/data owner	Artifact-level use/redistribution decision, de-identification review, public-demo bundle approval, retention and derivative access handling	Sharing affected assets. A checksum, generic collection license or research disclaimer is not sufficient authorization.
D04 external inference - project/AI owner	Account/model identity, permitted minimized payload, provider retention/training policy and tested grounding; SPEC-010 pp4-5	External transmission and generated prose publication. Keep adapter disabled and use correctly labeled approved/deterministic summaries until closed.
D05 scientific review and evaluation - engineering/scientific owner	Frozen suite and actual results, physical checks, original-patient split audit, failure/abstention counts and review evidence; SPEC-011	Full-readiness or model-performance claims. Missing mandatory results are not a pass; no arbitrary Dice or confidence threshold.
D06 deployment/demo feasibility - engineering/project owner	Target-device memory/mesh benchmark, restore/retry/access tests, chosen hosting configuration; official event deadline, rubric and required integrations	Final deployment/presentation commitments. Ten days and five minutes are planning assumptions, not verified event constraints.

Critical release blockers are cross-patient/branch leakage, synthetic-as-real labeling, invalid physical geometry, unauthorized exposure and unsupported clinical claims. Major blockers include stale UI state mixtures, partially published branches and absent real follow-up evidence. Disable failing optional capabilities rather than conceal failures; do not claim the full P0 prototype complete when a mandatory gate is open.

5 Coding agent handoff and approval record

1. Freeze the twelve attachment identities in a repository manifest with SHA-256 hashes and adopt this reconciliation overlay as COS-AUD-001 baseline 1.0. No source PDF rewriting is required. Record project-owner adoption before calling the baseline approved; that decision is not evidenced by this audit.

2. Begin repository/bootstrap, canonical types, synthetic engineering fixtures and database/authorization tests now. First implementation work closes C01-C04 and records C05-C06; do not defer API reconciliation to the end of the build.

3. Produce a single reviewed OpenAPI contract, shared generated client, numbered migrations and versioned config fixtures. These are implementation artifacts, not twelve replacement documents. Agents must not invent missing clinical fields, datasets, model entitlements or endpoint variants.

4. Build one real-data vertical slice only after D01/D03 close, then prove D02 early. Complete scenarios, updates and grounded interpretation against the same baseline. Keep learned forecasting, automated segmentation and new literature experiments outside the critical path.

5. Maintain a compact decision/gate ledger with owner, status, evidence and affected task IDs. Contract changes require a versioned decision and affected tests, not silent local agent reinterpretation. Use SPEC-011 results for release; SPEC-012 supplies task order, not permission to waive failures.

Final audit judgment: conditionally ready to start foundation implementation. Not ready to assert a fully approved, executable contract set or release-ready prototype until the adoption record, early contract closures and relevant evidence gates are completed. The architecture itself does not need replacement.